

SYNOPSIS

At about 0005 on 5 February 2013, a crewman from the motor tug *Endurance* fell overboard in rough seas about 2.3 miles west-south-west of Beachy Head on the south coast of England. The crewman fell while attempting to cross to the unmanned motor cruiser *Sirius M* with a replacement towline after the original towline connecting the two vessels had parted. The motor tug's skipper's efforts to recover the crewman back on board in rough seas were unsuccessful; the crewman soon lost consciousness and disappeared from view.

The skipper informed Dover coastguard of the situation and a search and rescue operation was immediately started. However, the crewman's body was not found until it came ashore under the cliffs of Beachy Head 11 weeks later.

The MAIB investigation identified that the attempt to reconnect a towline between *Endurance* and *Sirius M* was a desperate and ill-considered measure brought about by the use of poor towing practices, a disregard of the weather forecasts, and a lack of planning, risk assessment and emergency preparedness. Factors contributing to the accident included:

- *Endurance* was not certified to operate in the sea conditions experienced
- the skipper was not qualified to operate the vessel during the coastal sections of the intended voyage
- the skipper was not trained or qualified in towing operations and did not follow good practice
- decision-making and behaviour on board were likely to have been affected by fatigue
- safety was afforded a low priority on board
- the crewman was not wearing a lifejacket.

The investigation also identified that, although not directly contributory to this accident, the certification process conducted by the International Institute of Marine Surveyors, which allowed *Endurance* to operate out to sea, was not robust. Weaknesses were also identified in Medway Ports' compliance with the Port Marine Safety Code with respect to motor tug licensing and operation, accident investigation and the sharing of information.

Action has been taken by the International Institute of Marine Surveying, Medway Ports, Peel Ports Group and the owner/skipper of *Endurance* to address the safety issues identified. The Maritime and Coastguard Agency has also taken steps to ensure the lessons learned are shared with all the UK certifying authorities. A recommendation has been made to *Endurance*'s owner and skipper, which is intended to help ensure the safe operation of his vessel in the future.



Endurance

1.2 NARRATIVE

At about 0400 on 3 February 2013, the skipper of the motor tug¹ *Endurance* (**Figure 1**) and his crewman, Steven Trice met at Cuxton Marina near Rochester, England. They boarded a small boat and made their way to *Endurance's* River Medway mooring. *Endurance* was then manoeuvred across the river to Medway Bridge marina (**Figure 2**), where the skipper and Steven connected a towline from *Endurance* to the bow of the motor cruiser *Sirius M* (**Figure 3**). *Sirius M* was then towed by *Endurance* downriver. The vessels' destination was Brighton on the south coast of England. *Endurance's* skipper had checked the weather forecast using the *windfinder.com* website² and had assessed the predicted conditions to be suitable for the 2 day passage. He intended to berth the vessels overnight in Dover.

¹ The skipper was also the owner of the motor tug. Both titles will be used throughout the report. The Port of London Authority's definition of a motor tug is an inland waterways vessel which is less than or equal to 50gt and licensed to tow, push and manoeuvre small craft, barges and pontoons.

² Windfinder.com is a German based company that provides a global weather service for wind, wave and weather related outdoor sports. Its website combines weather forecasts for watersport locations and real time observations from more than 7000 weather stations worldwide.



Figure 3: *Sirius M* berthed at Medway Bridge marina

At 0513, the skipper informed Medway vessel traffic services (VTS), via very high frequency (VHF) radio, that *Endurance* was underway and was towing “a 60 foot steel cabin cruiser outward bound for the south coast”. The VTS officer acknowledged the call and replied:

“....you’ve got a 60 foot cruiser you say, outward bound for the south coast. We have been issued with a gale warning, a couple of hours ago. A south-westerly gale force 8 is expected soon³. Over”.

The skipper acknowledged the weather warning and he also told the VTS officer that he would “*probably chuck in at Ramsgate then*”. At 0641, Steven sent a text message to his girlfriend advising her that they were outward bound at Kingsnorth and that force 8 winds had been forecast.

At about 0730, *Endurance* passed the Medway VTS tower (**Figure 4**). Several minutes later, Medway VTS repeated the gale warning. The warning was again acknowledged by the skipper when he reported his position. He also repeated his intention to proceed to Ramsgate and to assess the situation from there. At 0826, Steven sent a text message to his girlfriend telling her that she could follow *Endurance’s* progress on the MarineTraffic.com⁴ website.

³ On the Beaufort wind scale ‘gale force 8’ winds are between 34 and 40kts. The Met Office definition of ‘soon’ is within 6 to 12 hours.

⁴ The MarineTraffic.com website is part of an open, community-based project, which is dedicated to collecting and presenting data which are exploited for research purposes. It provides free real-time information to the public about ship movements and ports. Its initial data collection is based on the signals received from maritime vessels’ Automatic Identification System (AIS) transmitters (see footnote 8).

As *Endurance* continued on an easterly heading, the tug made good a speed over the ground (SOG) of between 6 and 7 knots (kts), assisted by the ebb tide. At about 1200, *Endurance*'s skipper altered the tug and tow's heading to the south, towards North Foreland (**Figure 5**). Two hours later, as the vessels approached Ramsgate, the skipper decided to continue on passage to Dover.

Shortly after, the direction of the tidal stream turned and the strength of the south-westerly wind steadily increased to force 6, with gusts up to force 8. As a result, *Endurance*'s SOG was reduced to between 1 and 2kts. At 1730, Steven's girlfriend sent him a text message:

"will take forever at that pace. AIS shows 2kts. You must be tired"

At 2130, *Endurance* and *Sirius M* entered Dover (**Figure 5**). *Endurance* was refuelled and the vessels were moored alongside. The skipper inspected the towline. He also looked at the weather forecast for the following day. The inshore waters forecast⁵ issued by the Met Office for the area included:

west to south-westerly winds, force 5 to 7, occasionally gale-force 8 with moderate to rough seas.

Overnight, the skipper slept on board *Endurance* and Steven slept on board *Sirius M*. At 0906 the following morning (4 February), *Endurance* sailed from Dover with *Sirius M* in tow (**Figure 6**). The skipper intended to assess the sea conditions once on passage. If the tow was handling well, he intended to continue to Brighton. If not, he intended to return to Dover.

Soon after leaving Dover, Steven recorded the movement of *Sirius M* in the moderate swell on his camera (**Figure 7**). The video footage taken showed that the towline did not have a catenary⁶ and was constantly snatching⁷. It also showed that the motor tug's aft deck was awash with water. During the remainder of the day, the vessels' SOG fluctuated between 2.5kts and 5kts depending on the direction of the tidal stream.

At about 2030, *Endurance* passed the Sovereign Light (**Figure 8**). By then, the weather conditions had worsened and both the skipper and Steven felt sea-sick. The men discussed aborting the passage and seeking refuge in Eastbourne. Despite being tired and finding the conditions uncomfortable, they both wanted to get the job done, so the skipper made the decision to press on to Brighton.

At 2300, *Endurance* passed south of Beachy Head. It was approaching slack water and the vessel was heading into a strong westerly wind; the SOG was 3kts. At 2337, the towline parted and *Endurance*'s SOG increased to 6kts. At about 2340, the skipper realised what had happened and reduced speed. *Sirius M* was not in sight so the skipper turned *Endurance* onto a reciprocal heading to look for the motor cruiser. Within 3 minutes, *Endurance*'s skipper saw *Sirius M*; it was stopped in the water, lying beam onto the wind and was unlit.

⁵ Inshore waters forecast – for coastal areas up to 12 miles offshore.

⁶ Catenary: a curve formed by a wire, rope or chain hanging freely from two points that are not in the same vertical line.

⁷ Snatching – the generation of large dynamic forces in a towline as it is stretched taught.

Steven went on deck and recovered the remains of the parted tow rope back on board. When he returned to the wheelhouse, the skipper manoeuvred *Endurance* head on to the waves close to *Sirius M*. The skipper and Steven then discussed how they could re-connect the tow and agreed that Steven would take another towline across to *Sirius M*.

Steven donned a high visibility work coat and went onto the deck to prepare another towline. The skipper initially left the wheelhouse to assist, but had to return to the helm to prevent *Endurance* from falling beam-on to the wind and the sea.

Steven removed a 32mm diameter rope from the wheelhouse roof and coiled it down on the aft deck. He secured one end of the rope to the towing post and then pulled the eye in the other end and several metres of the rope along the starboard side of the vessel to the foredeck (**Figure 9a**).

At about midnight, *Endurance*'s skipper manoeuvred the motor tug's bow towards *Sirius M*'s leeward (starboard) side, aiming for an open section of main deck guardrails amidships (**Figure 9b**). Meanwhile, Steven stood on the foredeck with the eye of the tow rope over his shoulder; he was steadying himself by holding onto the coaming on the inner edge of the wheelhouse roof (**Figure 10a**).

When the vessels were in close proximity, Steven leapt from the foredeck towards *Sirius M* (**Figure 10b**). Almost simultaneously, the skipper manoeuvred *Endurance* astern as *Sirius M* came down off the crest of a wave and lurched towards his vessel. The skipper initially thought that Steven had crossed successfully onto the motor cruiser. However, as the distance between the two vessels increased, he could no longer see Steven and realised that he must have fallen into the sea.

To prevent Steven being crushed between the two vessels, the skipper continued to manoeuvre *Endurance* astern until the distance to *Sirius M* was about 15m (**Figure 11a**). He then went out on deck and saw Steven floating motionless on the sea surface off the starboard bow; he was lying face up with the eye of the tow rope under his armpits.

The skipper heaved in on the tow rope but soon had to stop and return to the wheelhouse to again manoeuvre *Endurance* clear of *Sirius M*. Once the vessels were again clear of each other, the skipper went back on deck and continued to pull in on the tow rope as quickly as he could (**Figure 11b**). When Steven was within 10m of *Endurance*, a large wave carried him under the tug's stern and the tow rope became entangled within the vessel's tyre fenders. By the time the skipper had freed the tow rope from the tyres, Steven had slipped from its eye. The skipper ran to the port side of the vessel and saw Steven drift into the darkness (**11c**) and out of sight.

At 0012 (on 5 February 2013), the skipper called Dover coastguard on VHF radio and advised that he had a "Mayday situation; or Pan Pan". He reported that he was a "small tug and tow bound for Brighton Marina", and that he had lost his tow and his crewman was in the water. The coastguard acknowledged the call and asked the skipper to confirm his position. Initially, the skipper explained that *Endurance* carried a Class A Automatic Identification System (AIS)⁸ and was 5 miles west of Beachy

⁸ AIS is an automatic tracking system used on ships and by VTS for identifying and locating vessels by electronically exchanging data with other nearby ships, AIS base stations, and satellites.

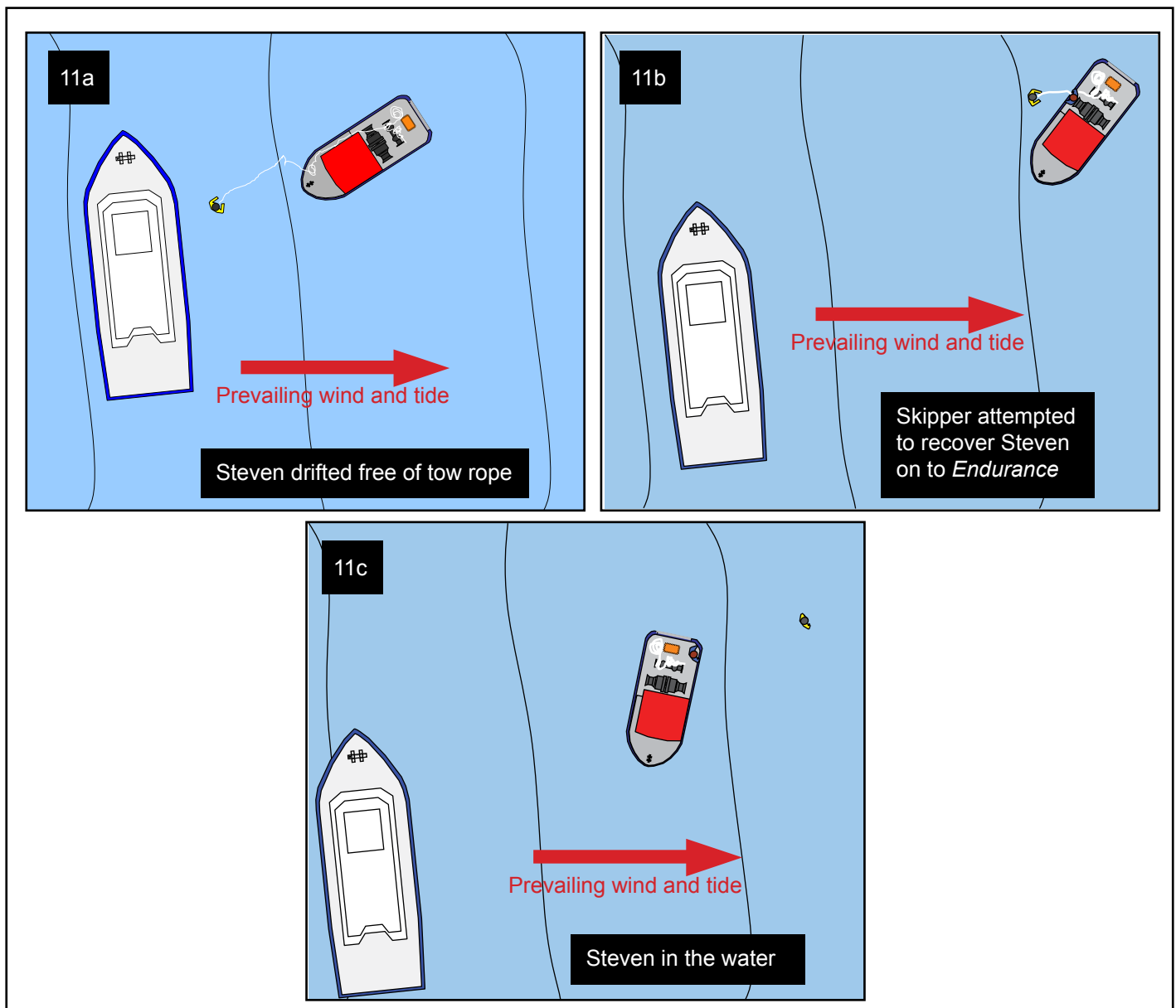


Figure 11a, 11b and 11c: Attempted recovery of Steven from the water

Head. About 1 minute later, he gave his position as 50° 43.729N and 000° 11.851E⁹, which was about 2.3 miles west-south-west of Beachy Head. When the skipper was asked by the coastguard operator whether his crewman was wearing any flotation aids, he replied:

“not quite sure, I think he was wearing one under his high-vis jacket, but I’m not quite sure”

The skipper also added that Steven was wearing jeans and a boiler suit under his jacket. In response to further questions, the skipper confirmed that *Sirius M* and *Endurance* were no longer connected and that *Sirius M* was unmanned and its navigation lights were not illuminated. He also advised that the sea conditions were moderate to rough.

⁹ The VHF radio exchange between *Endurance* and Dover Coastguard was monitored by the Marine Rescue and Co-ordination Centre (MRCC) Solent. As *Endurance* was within Solent’s area of responsibility, MRCC Solent immediately contacted Dover Coastguard to confirm which station would co-ordinate the search and rescue (SAR) operation and whether the Newhaven lifeboat should be tasked. Dover coastguard advised that as it was in contact with *Endurance*’s skipper, it would continue to co-ordinate the SAR and that it would probably initially task the Eastbourne lifeboat.

Between 0015 and 0020, Dover coastguard tasked a coastguard rescue helicopter (R104) and the Royal National Lifeboat Institution (RNLI) all-weather lifeboat (ALB) in Eastbourne to assist. The lifeboat arrived on scene at about 0059. At the same time, *Endurance*'s skipper informed the coastguard that he had not relocated *Sirius M* and that his crewman was probably not wearing a lifejacket because all of the vessel's lifejackets were still in the wheelhouse. He also explained that his crewman appeared to be unconscious when he last saw him.

At 0115, R104 arrived on scene and the Newhaven ALB was tasked to join the search. At about 0130, R104 located *Sirius M* drifting in an easterly direction in position 50° 44.04N 000° 17.19E. When the position of *Sirius M* was relayed to *Endurance*'s skipper by the coastguard, the skipper stated that he was operating single-handed and could not take the motor cruiser back under tow.

The coastguard was concerned about the wellbeing of *Endurance*'s skipper and requested the coxswain of the Eastbourne ALB to transfer one of his crew to the motor tug to assist. However, when the ALB arrived alongside *Endurance* its coxswain quickly decided that it was too dangerous to transfer crew due to the relative motion of the vessels in the rough seas and the large amount of seawater awash on the motor tug's aft deck.

At 0215, *Endurance*'s skipper requested the coastguard's permission to head into Eastbourne. By this time, the extent of the rolling and pitching of the tug was so severe that the skipper was concerned that it might lead to problems with the engine fuel supply. The coastguard approved the request and, at 0305, *Endurance* entered Eastbourne's Sovereign Harbour (**Figure 12**). *Sirius M* was later recovered by the Newhaven ALB and towed into Eastbourne (**Figure 13**).



Figure 12: *Endurance* entering Eastbourne's Sovereign Harbour

Almost 2 years later, *Endurance*'s skipper applied to Medway Ports for a licence to operate his vessel as a tug within its port limits. On his application for the licence, the skipper stated that he had 20 years of experience on the River Medway and the River Thames. The licence (**Annex A**) was subsequently issued on 1 April 2011.

On 16 August 2011, a PLA surveyor inspected *Endurance* to assess the vessel's fitness for purpose as a motor tug. Many of the items listed on the 2009 pre-licensing inspection report had been addressed, but no stability calculations had been completed and no guardrails had been fitted. The inspection report dated 16 September 2011 (**Annex B**) listed the remedial works and the additional equipment required prior to a licence being issued. The list of remedial works did not include the PLA's earlier requirement to provide stability calculations and guardrails.

On 28 December 2011, the PLA surveyor returned on board *Endurance* and was satisfied that all the remedial items during the previous visit had been addressed. At the skipper's request, the surveyor also assessed *Endurance*'s suitability to carry passengers.

Following the inspection, the PLA issued a licence which allowed the skipper to operate *Endurance* as a motor tug within its Category C²⁰ and Category D²¹ waters. It also provided a list of further remedial items that needed to be addressed before a passenger boat licence could be issued. These included:

- the completion of a heel test in accordance with Maritime and Coastguard Agency (MCA) guidelines²²
- the fitting of a secondary steering system for use in an emergency
- the implementation of a requirement for lifejackets to be worn at all times by all persons on deck
- the development of a safety management system.

A heel test was subsequently conducted on the vessel at Cuxton Marina on 17 January 2012 by an independent surveyor. However, *Endurance*'s skipper decided not to continue with his application to the PLA for approval to operate *Endurance* as a passenger vessel.

1.6.2 Seagoing voyages

During the heel test on 17 January 2012, *Endurance*'s skipper instructed the attending surveyor, who was registered as an authorised person with the IIMS certifying authority (CA), to examine the vessel with a view to certifying *Endurance* to operate at sea as a passenger carrying workboat. Accordingly, the surveyor examined *Endurance* against the requirements of the Small Commercial Vessel and

²⁰ Category C waters - tidal rivers, estuaries and large, deep lakes and lochs where the significant wave height could not be expected to exceed 1.2m at any time.

²¹ Category D waters - tidal rivers and estuaries where the significant wave height could not be expected to exceed 2m at any time.

²² For details see **paragraph 1.13.1**

Pilot Boat (SCV) Code²³. As part of the certification process, the CA surveyor also conducted an out of water inspection of the vessel at Denton Wharf, Gravesend during April 2012 (**Figure 16**).

On 2 August 2012, the surveyor submitted a completed *Application for Examination* form (SCV1) (**Annex C**), a *Compliance Examination and Declaration* report form (SCV2) (**Annex D**) and the heel test report (**Annex E**) to the IIMS CA committee for scrutiny. The paperwork forwarded to IIMS contained many clerical errors and factual discrepancies, including:

- The SCV1 and SCV2 forms and the heel test report were dated 28 June 2012.
- The owner's declarations on the SCV1 and SCV2 forms had not been signed by the owner.
- The initial vessel examination and out of water survey were incorrectly recorded as having been carried out in Newhaven and Cuxton respectively.
- The minimum required freeboard detailed on the SCV2 form was not correct for the vessel's length overall (LOA), and the freeboard measurements did not match the measurements recorded on the surveyor's heel test report.
- The name of *Endurance*'s owner detailed on the SCV2 form was incorrect.

The surveyor later re-submitted a revised SCV2 declaration sheet which corrected the location of the vessel examination and also added:

Code Section 22.2.3.4 Bulwarks with an aft railing to transom 600 will impede operation as a buoy layer and service. All passengers and crew to wear harnesses when on deck. [sic]

Again, the name of the vessel owner was incorrect and the owner did not sign the declaration.

The paperwork submitted by the surveyor was scrutinised by IIMS but was not challenged. On 31 August 2012, the CA certified *Endurance* to be operated commercially and carry a maximum of eight passengers out to sea up to area category 2²⁴ with the following restrictions:

- to only operate the vessel in favourable weather²⁵
- to carry no more than 3 persons on board when working at sea for over 24 hours.

²³ The SCV Code was published in 2004 by the MCA as an annex to Marine Guidance Note (MGN) 280 (M): *Small Vessels in Commercial Use for Sport or Pleasure, Workboats and Pilot Boats – Alternative Construction Standards*. The SCV Code (frequently referred to as the 'harmonised SCV Code') was intended to supersede the existing codes of practice (Blue – Small Commercial Sailing Vessels, Yellow – Small Commercial Motor Vessels, Brown – Small Workboats and Pilot Boats, and Red – Small Vessels in Commercial use for Sport or Pleasure Operating from a Nominated Point of Departure). The SCV Code has not yet been enabled but it is accepted as an equivalent alternative standard to the existing codes of practice.

²⁴ Area category 2 – up to 60 miles from a safe haven.

²⁵ Definition of favourable weather given in MGN 280(M) - *wind, sea and visibility conditions which are deemed by the skipper to be safe for a small vessel to operate within the limits applied to it; or, in any other case means conditions existing throughout a voyage or excursion in which the effects either individually or in combination of swell, height of waves, strength of wind and visibility cause no hazard to the safety of the vessel, including handling ability.*

On 7 December 2012, *Endurance*'s skipper submitted his annual self-certification survey report to the PLA in accordance with its licensing requirements. In the comments section of his survey report, the owner advised the PLA that *Endurance* had completed a heel test and was now coded as a *Category 2 SCV and could carry 10 passengers*.

1.7 POST-ACCIDENT SURVEYS AND INSPECTIONS

1.7.1 Towing arrangement

Examination of the parted towline indicated that the rope was old and in poor condition. MAIB inspectors reconstructed the towing eye in situ (**Figure 24**), and it was evident that the rope had parted at a point where it had been in contact with the steel rim of one of the motor cruiser's fairleads. The rope was also damaged where it had been in contact with the other fairlead (**Figure 25**). Paint on the forward edge of both fairleads had been removed and the base steel had a freshly polished



Figure 24: Tow rope securing arrangement reconstructed on *Sirius M*'s foredeck



Figure 25: Tow rope chafe marks and frayed ends, and fraying at the bowline knot used to form the towing eye

appearance. Blue paint could be seen on the rope in the areas where it had rubbed against the steel rim of the fairleads. It was also noted that the length of rope within and around the bowline had started to fray and wear.

1.7.2 Maritime and Coastguard Agency

On 6 February 2013, the MCA inspected *Endurance* in Eastbourne. During the inspection, the MCA surveyor identified 12 deficiencies, including:

- the skipper's BML was not an adequate qualification for the voyage undertaken

- no safety harnesses were carried on board (minimum of 2 required)
- no paper charts or almanacs for the passage were carried on board
- no man overboard recovery ladder or net were carried on board
- the liferaft was overdue for service
- the engine room bilge alarm was not working.

The MCA surveyor also identified that the VHF radio's DSC unit was not connected to the GPS. The MCA informed the IIMS of the accident and of the findings of its inspection. As a result, the IIMS suspended *Endurance* from its CA Register.

1.7.3 International Institute of Marine Surveying

On 13 February 2013, *Endurance* was inspected by IIMS surveyors to assess the vessel's compliance with the SCV Code. The inspection established that, apart from the inadequacy of the skipper's qualification for the voyage undertaken, the deficiencies identified during the MCA's general inspection had been rectified. It also identified a number of other significant deficiencies, including:

- The emergency steering could only be accessed through the flush deck watertight hatch aft of the main weather deck winch (**Figure 26**).
- The hydraulic steering system had not been fitted with a bypass valve and therefore could not be operated manually in an emergency.
- No jackstays had been provided for use with safety harnesses.
- Radio procedures cards had not been provided.

A review by IIMS of the heel test report submitted in 2012 also identified several anomalies. In particular, the freeboard measurements were unusually uniform and the minimum required freeboard had not been calculated. Consequently, the IIMS surveyors concluded that *Endurance* should not have passed the simplified stability assessment.

1.7.4 Heel test and freeboard measurement

On 29 April 2013, *Endurance* was subjected to a further heel test and freeboard measurement. This was undertaken by an independent surveyor at the request of the surveyor who conducted the initial stability assessment in January 2012. The resulting test report concluded that *Endurance* did not have the freeboard required by the SCV Code to operate at sea.

1.8 MOTOR TUG LICENSING REQUIREMENTS

Tugs operating within the PLA's harbour limits are licensed by its licensing department, which is capable of conducting its own vessel surveys and inspections. Vessels surveyed and subsequently licensed to operate as motor tugs by the

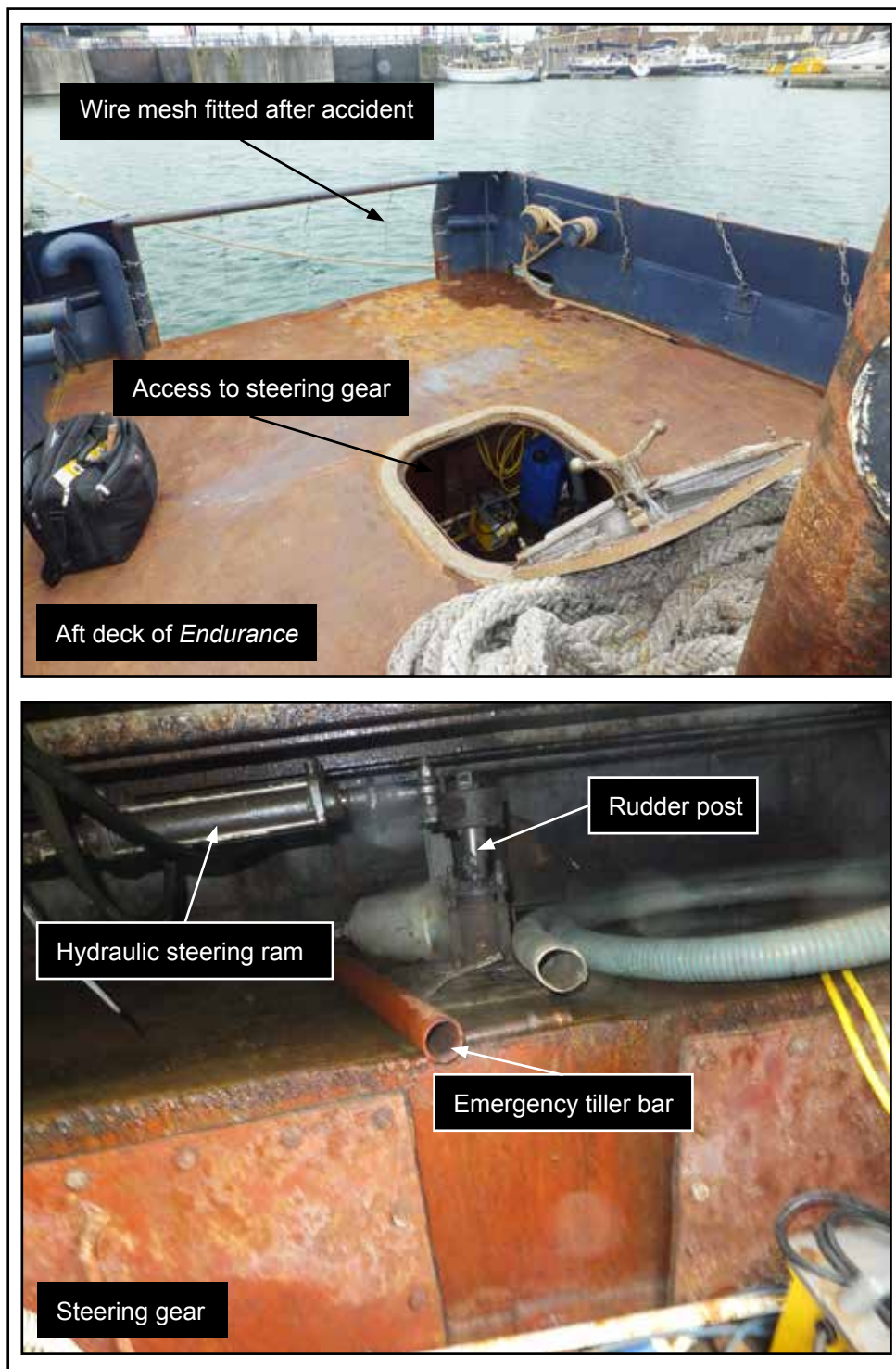


Figure 26: *Endurance's* steering gear arrangement

PLA are generally accepted and licensed by Medway Ports for operation on the River Medway without additional inspection, following the payment of a small administration charge.

The terms and conditions set out in Medway Port's tug licence (**Annex A**) require owners (or managers) to ensure that their vessels are:

- manned by a competent master and crew

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 OVERVIEW

Steven Trice died because he fell into the sea while trying to transfer from *Endurance*'s foredeck to the deck of *Sirius M* and his skipper was unable to recover him back on board. Steven's attempt to transfer from *Endurance* to *Sirius M* when both vessels were underway in rough seas and in darkness was extraordinarily dangerous. It was a desperate and ill-considered measure brought about by the use of poor towing practices, a disregard of the weather forecasts, and a lack of planning, risk assessment and emergency preparedness. In addition, *Endurance* was not certificated to operate in the sea conditions experienced and the skipper was not qualified to undertake the coastal part of the intended voyage.

2.3 EMERGENCY RESPONSE

2.3.1 Attempted recovery of the tow

By the time *Endurance*'s skipper realised the towline had parted, *Sirius M* was no longer in sight. The vessel's stern light was probably not lit because it had either not been switched on when the vessels left Dover, or the battery supplying its power had run out of charge. In the darkness, the unlit and unmanned motor cruiser was a potential danger to other vessels. However, *Endurance*'s skipper did not alert the coastguard to the situation and request assistance.

This was a significant omission given that the skipper had not rigged an emergency towline. Therefore, he had no means of re-securing the tow other than physically taking a replacement towline across to *Sirius M*. Although *Endurance*'s skipper discussed the situation with Steven, it is clear from the motor tug's approach to *Sirius M*, and the failure to take any precautions, that neither of the men appreciated or considered the considerable risks involved.

Sirius M was lying beam-on to the wind and waves. In such situations it is usual practice for a rescue vessel to approach a disabled vessel by passing close by its bow or stern (crossing the 'T') to avoid the leeward danger zone (**Figure 29**). In this case, *Endurance* approached *Sirius M* on the drifting vessel's leeward side, aiming amidships (**Figure 9**). Heading into the wind and sea possibly made *Endurance* a more stable platform, and the motor cruiser's superstructure would have provided some shelter. Nonetheless, the risk of *Sirius M* being set towards *Endurance* when the vessels were close was significant and predictable.

In view of the danger caused by the relative movement of the vessels in the rough seas, the darkness, and the difficulty in carrying the towline, it is clear that Steven's transfer across to *Sirius M* should not have been attempted. This is endorsed by the fact that during the SAR the coxswain of the Eastbourne ALB would not transfer one

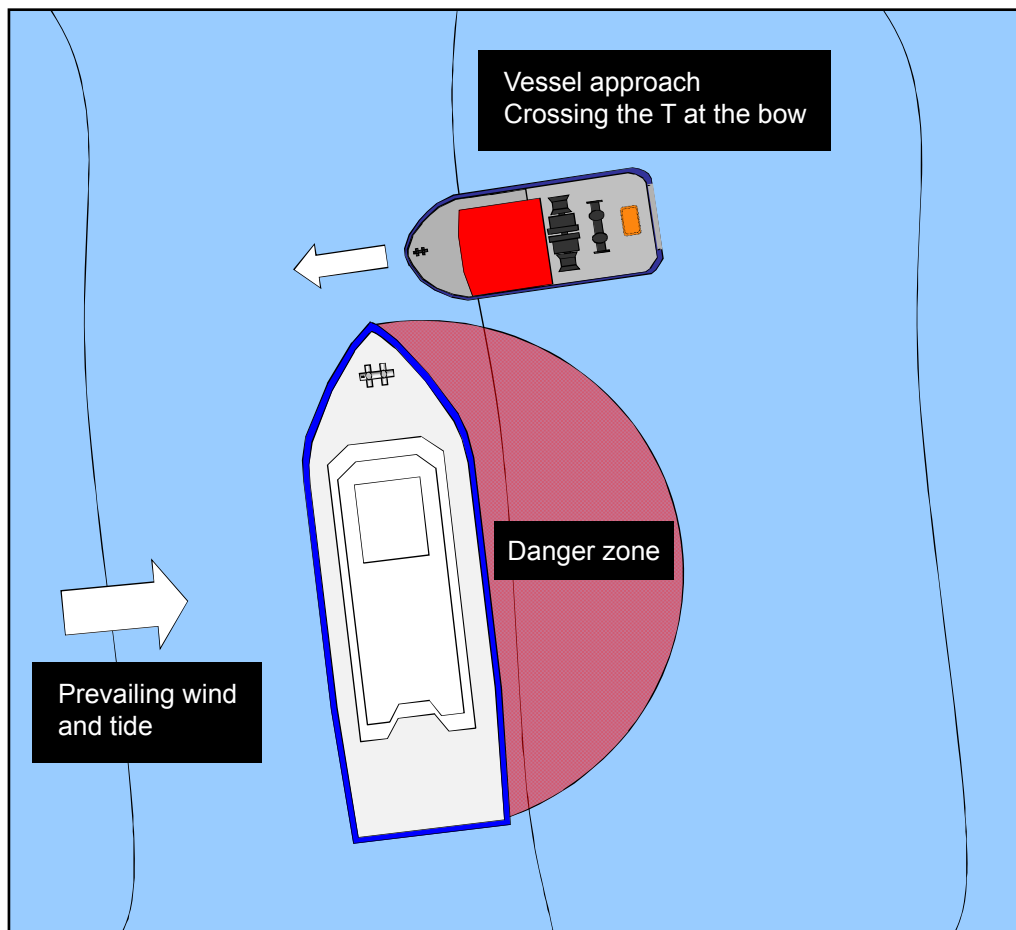


Figure 29: Crossing the 'T'

of his crew across to *Endurance* because it was not safe to do so. Given Steven's concern over the voyage, it is puzzling why he did not don his flotation suit and a lifejacket when faced with such a perilous task.

2.3.2 Man overboard and attempted recovery

It is not certain why Steven fell into the water. It is possible that he was impeded by the towline or that he misjudged the distance between the two vessels or their relative motions. Equally, he might have slipped or lost his grip on either vessel as he attempted to step or jump from one to the other.

The skipper's initial reaction of manoeuvring *Endurance* away from *Sirius M* when he realised that Steven had probably fallen into the sea, was positive. However, the skipper did not immediately alert the coastguard to the situation, which he could have easily done by pressing the DSC distress button before leaving the wheelhouse. He also did not throw a lifebuoy into the water, which might have assisted in the subsequent search.

In the rough sea conditions, the darkness and the continuing close proximity of *Sirius M*, it would have been virtually impossible for the skipper to recover Steven without assistance. Indeed, without a man overboard recovery device, it would have been difficult for the skipper to recover any person falling overboard from *Endurance* by himself, even in calm waters.

Initially, Steven was attached to the eye of the replacement towline, but the skipper had to divide his efforts between pulling him towards *Endurance* and avoiding *Sirius M*. As soon as Steven lost consciousness and slipped from the eye of the towline, the skipper had few options remaining; he was alone and had no means of recovering Steven back on board. It was only then that he alerted the coastguard to the man overboard and the lost tow.

Although the postmortem report was unable to establish the exact cause of Steven's death, it is most likely that he drowned. The sea temperature was less than 9°C and it is likely that Steven would have suffered from the effects of cold water shock to some degree³⁶. Survival times in rough seas without the support of a PFD or other buoyant object can typically be measured in minutes rather than hours. Steven's chances of survival were reduced considerably by him not wearing a lifejacket or his insulated flotation suit.

2.4 FATIGUE

Although the skipper and Steven had the opportunity to sleep well overnight in Dover, both men were likely to have been feeling tired as *Endurance* passed south of Beachy Head (**Figure 8**). By then, *Endurance* had been underway in heavy sea conditions for over 14 hours, and both of the men were sea sick. In addition, the skipper had spent about 32 of the previous 44 hours at the helm while the vessel was underway because Steven was not confident steering *Endurance* at sea while towing. Taking into consideration that the skipper had completed a near 10 hour shift as a deckhand on the River Thames only 2 hours before setting off from Cuxton Marina, his decision-making and behaviour were likely to have been affected by fatigue to some degree.

2.5 REGULATORY COMPLIANCE

Endurance was certified to operate as a motor tug on the River Medway and the River Thames by Medway Ports and the PLA respectively. The vessel was also certified, albeit erroneously, by the IIMS to operate under the SCV Code as a workboat up to 60 miles from a safe haven in favourable weather. As the sea state at the time of the accident was moderate to rough, with a 2m swell, the conditions were clearly outside the parameters of 'favourable weather' detailed in the SCV Code and the MCA requirements for towage operations. Indeed, during the SAR, the rolling was so severe that *Endurance's* skipper became concerned about losing the fuel supply to the main engine. The vessel should not have been at sea in such conditions.

In addition, *Endurance's* skipper's BML only allowed him to operate out to 3 miles from land and up to 15 miles from a point of departure. Therefore, he was not qualified to skipper his vessel during the intended passage between Dover and Brighton. Moreover, as the skipper had not added a 'towing and pushing' endorsement to his BML, he was not qualified to operate *Endurance* as a motor tug on the River Medway or the River Thames. He should also not have taken *Endurance* in and out of Dover without a local knowledge endorsement for the port.

³⁶ Cold water shock can occur following sudden immersion in water whose temperature is 15°C or below. The cold can paralyze muscles, cause muscle spasms, and a rise in heart rate and blood pressure resulting in a heart attack. The spasms and a gasp reflex can cause water to be ingested or for the breath to be held involuntarily.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The attempt to reconnect a towline between *Endurance* and *Sirius M* was a desperate and ill-considered measure brought about by the use of poor towing practices, a disregard of the weather forecasts, and a lack of planning, risk assessment and emergency preparedness. [2.2]
2. Neither the skipper nor the crewman appreciated or considered the considerable risks of the crewman transferring across to the unmanned tow. [2.3.1]
3. The crewman's chances of survival were reduced considerably by him not wearing a lifejacket or his insulated flotation suit. [2.3.2]
4. The skipper's decision-making and behaviour are likely to have been affected by fatigue to some degree. [2.4]
5. In accordance with the limitations of its certification, *Endurance* should not have been operating at sea in the conditions experienced. [2.5]
6. The skipper was not qualified to operate *Endurance* during the intended passage between Dover and Brighton. [2.5]
7. The tow rope parted due to chafe abrasion caused by the rubbing action of the rope against the steel rims of *Sirius M*'s bulwark fairleads. This undoubtedly occurred because the arrangements did not follow good practice in safety-critical areas. [2.6]
8. The skipper's apparent lack of competency in towing operations at sea reflected his lack of formal training in this sphere of work. [2.7]
9. *Endurance*'s skipper gave a low priority to safety both during the passage to Brighton and during his more usual operations on the River Medway and the River Thames. [2.8]
10. *Endurance*'s skipper referred to unofficial sources of weather information that were potentially misleading. [2.9]

3.2 OTHER SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT

1. When the crewman fell into the water, the skipper did not immediately alert the coastguard to the situation or throw a lifebuoy into the water. [2.3.2]

3.3 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The skipper had not added a 'towing and pushing' endorsement to his BML and was therefore not qualified to operate *Endurance* as a motor tug on the River Medway or the River Thames. [2.5]
2. The examinations and heel test conducted by IIMS in 2012, which enabled